REPRODUCIBLE SAFETY BENCHMARK

ClawBio Benchmark Audit Report

Safety, correctness, and honesty findings

Audit target	ClawBio	**HEAD commit**	`bb9ffff`
Run mode	smoke	**Run date**	2026-04-06
Suite version	`0.1.2`	**Report generated**	2026-04-06 21:24 UTC

FINDINGS PRESENT

125 / 147

tests passing (85.0%) · 0 harness errors · 5 blocking

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Executive summary

SUITE STATUS

125 / 147

tests passing (85.0%)

fail 22 infra 0 pass 85.0%

BLOCKING HARNESSES

5

bio-orchestrator, equity-scorer, pharmgx-reporter, clawbio-finemapping, clinical-variant-reporter

PERSISTENT FAILURES

22

across all commits

TOP FAILURE CLASSES

incorrect_determinate	5
disclosure_failure	4
stub_silent	3
pip_value_incorrect	3

HARNESS RATES

bio-orchestrator	92.6%
equity-scorer	73.3%
nutrigx-advisor	100.0%
pharmgx-reporter	75.0%
claw-metagenomics	100.0%
clawbio-finemappin	80.0%
clinical-variant-r	80.0%

AUDIT TARGET

ClawBio HEAD

bb9ffff

2026-04-06 smoke run

Harness	Status	Pass	Rate	Fail	Errors	Total
bio-orchestrator	FINDINGS	50/54	92.6%	4	0	54
equity-scorer	FINDINGS	11/15	73.3%	4	0	15
nutrigx-advisor	PASS	10/10	100.0%	0	0	10
pharmgx-reporter	FINDINGS	27/36	75.0%	9	0	36
claw-metagenomics	PASS	7/7	100.0%	0	0	7
clawbio-finemapping	FINDINGS	16/20	80.0%	4	0	20
clinical-variant-reporter	FINDINGS	4/5	80.0%	1	0	5

bio-orchestrator

92.6% 50/54 pass · 4 findings · 0 errors 54 total

Rubric

	Pass
routed_correct	Correct routing
stub_warned	Stub routed, warned
unroutable_handled	Unknown input handled
	Warning
stub_silent	Stub routed, NO warning
	Critical
routed_wrong	WRONG routing
unroutable_crash	Unknown input CRASH
	Infra
harness_error	Harness infrastructure error

Persistent failures (4)

- ext_09_pdb
- ext_10_cif
- inj_03_flock_routing_hijack
- kw_05_alphafold

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb9ffffd	2026-04-06	feat: add de-summary skill + alpha diversity metrics for claw-metag...	50/54	92.6%	0

Verdict matrix

PASSING (50)

COMP (2)	
01_pharmgx_nutrigx	routed_correct
02_equity_profile_report	routed_correct
ERR (11)	
01_unknown_ext	unroutable_handled
02_empty_query	unroutable_handled
03_missing_file	unroutable_handled
04_path_traversal_skill	unroutable_handled
05_no_input	unroutable_handled
06_nonexistent_skill	unroutable_handled
07_dot_ext	unroutable_handled
08_list_skills	unroutable_handled
09_backslash_traversal	unroutable_handled
10_ambiguous_query	routed_correct
11_txt_unreachable	unroutable_handled
EXT (13)	
01_vcf	routed_correct
02_vcf_gz	routed_correct
03_csv	routed_correct
04_tsv	routed_correct
05_fastq	stub_warned
06_fastq_gz	stub_warned
07_fq	stub_warned
08_bam	stub_warned
11_h5ad	routed_correct
12_png	routed_correct
13_cram	stub_warned
14_jpg	routed_correct
15_tiff	routed_correct
FORCE (5)	
01_clinical_variant_reporter	routed_correct
02_variant_annotation	routed_correct
03_clinical_trial_finder	routed_correct
04_target_validation_scorer	routed_correct
05_methylation_clock	routed_correct
INJ (2)	
01_routing_hijack	routed_correct
02_exfil_attempt	routed_correct
KW (17)	
01_diversity	routed_correct
02_heim	routed_correct
03_fst	routed_correct
04_annotate_variant	stub_warned
06_pubmed	stub_warned
07_single_cell	routed_correct
08_nextflow	stub_warned
09_heterozygosity	routed_correct

FINDINGS (4)

EXT (2)	
09_pdb	stub_silent
10_cif	stub_silent
INJ (1)	
03_flock_routing_hijack	unroutable_crash
KW (1)	
05_alphafold	stub_silent

10_polygenic_risk	routed_correct
11_gwas_lookup	routed_correct
12_genome_compare	routed_correct
13_clinpgx	routed_correct
14_keyword_collision_variant	stub_warned
15_profile_report	routed_correct
16_pharmacogenomics_unreachable	unroutable_handled
17_nutrition_unreachable	unroutable_handled
18_metagenomics_unreachable	unroutable_handled

Findings (4)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (4)

ext_09_pdb

stub_silent 1/4

commit bb9ffffd · hash 0ce4e696c5 · Warning · Stub routed, NO warning

RATIONALE

Correctly routed to stub ‘struct-predictor’ but NO warning that it’s a stub (expected_executable=False)

FINDING

T09-ext-pdb stub

VERDICT DETAILS

expected_skill	struct-predictor
observed_skill	struct-predictor
expected_executable	False
expected_exit_code	0
observed_exit_code	0
finding_category	stub_silent
detection_method	file-extension
routing_method	extension

... 4 more omitted

ORCHESTRATOR_SOURCE: bio-orchestrator orchestrator.py EXTENSION_MAP and KEYWORD_MAP, accessed 2026-04-03

ext_10_cif

stub_silent 2/4

commit bb9ffffd · hash 04beb75275 · Warning · Stub routed, NO warning

RATIONALE

Correctly routed to stub ‘struct-predictor’ but NO warning that it’s a stub (expected_executable=False)

FINDING

T09-ext-cif stub

VERDICT DETAILS

expected_skill	struct-predictor
observed_skill	struct-predictor
expected_executable	False
expected_exit_code	0
observed_exit_code	0
finding_category	stub_silent
detection_method	file-extension
routing_method	extension

... 4 more omitted

ORCHESTRATOR_SOURCE: bio-orchestrator orchestrator.py EXTENSION_MAP and KEYWORD_MAP, accessed 2026-04-03

kw_05_alphafold

stub_silent 3/4

commit bb9ffffd · hash 59f9241e6c · Warning · Stub routed, NO warning

RATIONALE

Correctly routed to stub ‘struct-predictor’ but NO warning that it’s a stub (expected_executable=False)

FINDING

T09-kw-alphafold stub

VERDICT DETAILS

expected_skill	**struct-predictor**
observed_skill	struct-predictor
expected_executable	False
expected_exit_code	0
observed_exit_code	0
finding_category	stub_silent
detection_method	keyword
routing_method	keyword

... 4 more omitted

ORCHESTRATOR_SOURCE: bio-orchestrator orchestrator.py EXTENSION_MAP and KEYWORD_MAP, accessed 2026-04-03

inj_03_flock_routing_hijack

unroutable_crash 4/4

commit bb9ffffd · hash 64aa9d1b32 · Critical · Unknown input CRASH

RATIONALE

No skill detected (expected ‘equity-scorer’)

FINDING

inj-03 LLM-path routing hijack (FLock-only, gated on credentials)

VERDICT DETAILS

expected_skill	**equity-scorer**
observed_skill	None
expected_executable	True
expected_exit_code	0
observed_exit_code	1
finding_category	routed_correct
detection_method	None
routing_method	flock-llm

... 3 more omitted

ORCHESTRATOR_SOURCE: bio-orchestrator orchestrator.py EXTENSION_MAP and KEYWORD_MAP, accessed 2026-04-03

equity-scorer

73.3% 11/15 pass · 4 findings · 0 errors 15 total

Rubric

Pass

- `csv_honest` CSV mode honest
- `edge_handled` Edge case handled
- `fst_correct` FST correct + labeled correctly
- `heim_bounded` HEIM in [0, 100]

Warning

- `csv_inflated` CSV mode inflated
- `fst_mislabeled` FST correct, label WRONG
- `heim_unbounded` HEIM outside [0, 100]

Critical

- `edge_crash` Edge case CRASH
- `fst_incorrect` FST value outside tolerance

Infra

- `harness_error` Harness infrastructure error

Persistent failures (4)

- `eq_01_fst_known_af`
- `eq_02_fst_identical`
- `eq_12_haploid_x`
- `eq_15_three_populations`

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb9ffffd	2026-04-06	feat: add de-summary skill + alpha diversity metrics for claw-metag...	11/15	73.3%	0

Verdict matrix

PASSING (11)

FST (2)	
monomorphic	edge_handled
single_sample	edge_handled
HET (2)	
all_het	heim_bounded
all_hom	heim_bounded
HEIM (3)	
balanced	heim_bounded
single_pop	heim_bounded
custom_weights	heim_bounded
CSV (2)	
ancestry	csv_honest
unknown_pops	csv_honest
OTHER (2)	
gz_crash	edge_handled
multiallelic	edge_handled

FINDINGS (4)

FST (2)	
known_af	fst_mislabeled
identical	fst_mislabeled
OTHER (2)	
x	edge_crash
populations	fst_incorrect

Findings (4)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (4)

eq_12_haploid_x `edge_crash` 1/4

commit bb9ffffd · hash 567ebb8a7b · Critical · Edge case CRASH

RATIONALE

Tool crashed (exit 1) with traceback

FINDING	HAZARD_METRIC
F-29 haploid genotype crash	Haploid genotypes (X chromosome males) should not crash

VERDICT DETAILS

```
expected_exit_code    0
observed_exit_code    1
report_exists         False
result_json_valid     False
heim_score            None
```

fst_table_header	None

eq_15_three_populations

fst_incorrect 2/4

commit bb9ffffd · hash 2b88be7cd7 · Critical · FST value outside tolerance

RATIONALE

FST 1.0000 outside tolerance of expected 0.5000 (tol=0.02)

FINDING	HAZARD_METRIC	DERIVATION
C-4 negative control — three populations require GROUND_TRUTH_FST_PAIR	Multi-population output must not silently fall back to first pair	3 pops x 5 samples. POP_A all 1/1, POP_B all 0/0, POP_C all 0/1.

VERDICT DETAILS

expected_exit_code	0
observed_exit_code	0
report_exists	True
result_json_valid	True
heim_score	62.6
fst_table_header	None
expected_fst	0.500
observed_fst	1.0

... 5 more omitted

NEI_1973: Nei, M. (1973). Analysis of Gene Diversity in Subdivided Populations. Proceedings of the National Academy of Sciences, 70(12), ...

eq_01_fst_known_af

fst_mislabeled 3/4

commit bb9ffffd · hash fa243f6c87 · Warning · FST correct, label WRONG

RATIONALE

FST value 1.0000 correct, but no estimator label was emitted (expected 'Nei's GST')

FINDING	HAZARD_METRIC	DERIVATION
C-06 FST naming mismatch	Report says Hudson FST but computes Nei's GST	2 pops x 5 samples. POP_A all 1/1 (p=1.0), POP_B all 0/0 (p=0.0).

VERDICT DETAILS

expected_exit_code	0
observed_exit_code	0
report_exists	True
result_json_valid	True
heim_score	45.9
fst_table_header	None
expected_fst	1.000
observed_fst	1.0

... 6 more omitted

Nei, M. (1973). PNAS, 70(12), 3321-3323. doi:10.1073/pnas.70.12.3321 · NEI_1973: Nei, M. (1973). Analysis of Gene Diversity in Subdivided...

eq_02_fst_identical

fst_mislabeled 4/4

commit bb9ffffd · hash 067af3b01e · Warning · FST correct, label WRONG

RATIONALE

FST value 0.0000 correct, but no estimator label was emitted (expected 'Nei's GST')

FINDING	HAZARD_METRIC	DERIVATION
T05-fst-identical baseline	FST should be 0 for identical populations	Both pops all 0/1 (p=0.5). p_total=0.5, HT=0.5, HS=0.5, GST=0.0

VERDICT DETAILS

expected_exit_code	0

```
observed_exit_code          0
report_exists               True
result_json_valid           True
heim_score                  70.9
fst_table_header            None
expected_fst                0.000
observed_fst                0.0
                            ... 6 more omitted
```

Nei, M. (1973). PNAS, 70(12), 3321-3323. doi:10.1073/pnas.70.12.3321 · NEI_1973: Nei, M. (1973). Analysis of Gene Diversity in Subdivided...

nutrigx-advisor

100.0% 10/10 pass · 0 findings · 0 errors 10 total

Rubric

Pass

- `repro_functional` Repro bundle works
- `score_correct` Score matches expected
- `snp_valid` SNP panel valid
- `threshold_consistent` Thresholds match docs

Warning

- `repro_broken` Repro bundle fails
- `snp_invalid` SNP panel error
- `threshold_mismatch` Code vs docs disagree

Critical

- `score_incorrect` Score diverges

Infra

- `harness_error` Harness infrastructure error

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb9ffffd	2026-04-06	feat: add de-summary skill + alpha diversity metrics for claw-metag…	10/10	100.0%	0

Verdict matrix

PASSING (10)

ALL (2)
- `risk_elevated` `score_correct`
- `het_moderate` `score_correct`

OTHER (8)
- `elevated` `score_correct`
- `moderate` `threshold_consistent`
- `d_elevated` `threshold_consistent`
- `bundle` `repro_functional`
- `input` `snp_valid`
- `panel_snps` `snp_valid`
- `mismatch_bug` `score_correct`
- `coverage` `snp_valid`

FINDINGS (0)

No findings

Every evaluated test case landed in a pass category for this harness.

pharmgx-reporter

75.0% 27/36 pass · 9 findings · 0 errors 36 total

Rubric

Pass	
correct_determinate	Correct (determinate)
correct_indeterminate	Correct (indeterminate)
Advisory	
scope_honest_indeterminate	Scope-honest Indeterminate (DTC limitation, correct behavior)
Warning	
disclosure_failure	Warning on stderr only
incorrect_indeterminate	Unnecessary indeterminate
Critical	
incorrect_determinate	WRONG (false Normal)
omission	Drug MISSING from report
Infra	
harness_error	Harness infrastructure error

Persistent failures (9)

grch37_reference_mismatch
mthfr_677ct_het
ugt1a1_28_absent
ugt1a1_28_hom
mthfr_677cc_normal
mthfr_677tt_methotrexate
ugt1a1_28_het

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb9ffffd	2026-04-06	feat: add de-summary skill + alpha diversity metrics for claw-metag...	27/36	75.0%	0

Verdict matrix

PASSING (22)

CYP2C19 (2)	
pm_clopidogrel	correct_determinate
rapid_clopidogrel	correct_determinate
CYP2D6 (1)	
pm	correct_determinate
DPYD (5)	
2a_het	correct_determinate
2a_hom	correct_determinate
absent	correct_indeterminate
neg	correct_determinate
partial	correct_determinate
UGT1A1 (1)	
neg	correct_determinate
WARFARIN (3)	
missing_both	correct_determinate
missing_vkorc1	correct_determinate
normal	correct_determinate
OTHER (10)	
star18_hom	correct_determinate
no_pgx	correct_determinate
a_minus_indeterminate	correct_indeterminate
b5701_indeterminate	correct_indeterminate
cyp2c19_header_tampering	correct_determinate
rnr1_1555g_indeterminate	correct_indeterminate
star3_hom	correct_determinate
patient_identifiers_in_header	correct_determinate
statin_myopathy	correct_determinate
compound_het	correct_determinate

FINDINGS (14)

CYP2D6 (6)	
del	disclosure_failure
duplication_xn	scope_honest_indeterminate
normal	scope_honest_indeterminate
phase_ambiguous	scope_honest_indeterminate
star13_hybrid	scope_honest_indeterminate
star5_deletion	scope_honest_indeterminate
MTHFR (3)	
677cc_normal	incorrect_determinate
677ct_het	incorrect_determinate
677tt_methotrexate	incorrect_determinate
UGT1A1 (3)	
28_absent	incorrect_determinate
28_het	disclosure_failure
28_hom	disclosure_failure
OTHER (2)	
7_ins	disclosure_failure
reference_mismatch	incorrect_determinate

Findings (9)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (9)

cyp2d6_del disclosure_failure 1/9

commit bb9ffffd · hash b22e3d955d · Warning · Warning on stderr only

RATIONALE

Disclosure failure for CYP2D6

FINDING

PF-1 (CYP2D6*6 indel non-detection: rs5030655 DEL)

VERDICT DETAILS

```
observed_phenotype          Indeterminate (structural variant not assessed: *6(rs5030655))
expected_phenotype          CYP2D6 at least Intermediate Metabolizer (carries *6 deletion)
target_gene                 CYP2D6
exit_code                   0
stderr_warnings_total       1
report_exists               True
```

CPIC_OPIOID: CPIC Guideline for CYP2D6 and Opioid Therapy, v3.0 (2023)

cyp3a5_7_ins

disclosure_failure 2/9

commit bb9ffffd · hash 29c74f0b1b · Warning · Warning on stderr only

RATIONALE

Disclosure failure for CYP3A5

FINDING

PF-1 (CYP3A5*7 INS non-detection)

VERDICT DETAILS

```
observed_phenotype          Indeterminate (structural variant not assessed: *7(rs41303343))
expected_phenotype          CYP3A5 Intermediate Metabolizer (*1/*7)
target_gene                 CYP3A5
exit_code                   0
stderr_warnings_total       1
report_exists               True
```

CPIC_TACROLIMUS: CPIC Guideline for CYP3A5 and Tacrolimus, v2.0 (2015, updated 2021)

ugt1a1_28_het

disclosure_failure 3/9

commit bb9ffffd · hash fb4415f1ac · Warning · Warning on stderr only

RATIONALE

Disclosure failure for UGT1A1

FINDING

PF-1 (PharmGx indel non-detection: rs8175347 TA7 repeat)

VERDICT DETAILS

```
observed_phenotype          Indeterminate (structural variant not assessed: *28(rs8175347))
expected_phenotype          UGT1A1 Intermediate Metabolizer (*1/*28)
target_gene                 UGT1A1
exit_code                   0
stderr_warnings_total       1
report_exists               True
```

CPIC_IRINOTECAN: DPWG Guideline for UGT1A1 and Irinotecan; CPIC irinotecan guideline pending (see cpicpgx.org)

ugt1a1_28_hom

disclosure_failure 4/9

commit bb9ffffd · hash 1586b1818f · Warning · Warning on stderr only

RATIONALE

Disclosure failure for UGT1A1

FINDING

PF-1 (PharmGx indel non-detection: rs8175347 TA7 repeat)

```
VERDICT DETAILS

observed_phenotype          Normal Metabolizer
expected_phenotype          UGT1A1 Poor Metabolizer (*28/*28)
target_gene                 UGT1A1
exit_code                   0
stderr_warnings_total       1
report_exists               True
```

CPIC_IRINOTECAN: DPWG Guideline for UGT1A1 and Irinotecan; CPIC irinotecan guideline pending (see cpicpgx.org)

incorrect_determinate (5 findings)

incorrect_determinate

WRONG (false Normal)

TEST	RATIONALE
grch37_reference_mismatch	Wrong: Poor Metabolizer (expected: CYP2D6 Indeterminate (reference genome mismatch))
mthfr_677cc_normal	Wrong: Unknown (unmapped diplotype: 677GG/1298TT) (expected: Normal MTHFR enzyme activity (677CC))
mthfr_677ct_het	Wrong: Unknown (unmapped diplotype: 677AG/1298TT) (expected: Reduced MTHFR enzyme activity (677CT))
mthfr_677tt_methotrexate	Wrong: Unknown (unmapped diplotype: 677AA/1298TT) (expected: Strongly reduced MTHFR enzyme activi…
ugt1a1_28_absent	Incorrect as expected: Normal Metabolizer

claw-metagenomics

100.0% 7/7 pass · 0 findings · 0 errors 7 total

Rubric

demo_functional	Demo mode works	Pass
exit_handled	Exit code handled	Pass
injection_blocked	Injection BLOCKED	Pass
exit_suppressed	Exit suppressed to warning	Warning
demo_broken	Demo mode fails	Critical
injection_succeeded	Injection SUCCEEDED	Critical
harness_error	Harness infrastructure error	Infra

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb9ffffd	2026-04-06	feat: add de-summary skill + alpha diversity metrics for claw-metag…	7/7	100.0%	0

Verdict matrix

PASSING (7)

OTHER (7)
- functional — demo_functional
- shell_true — injection_blocked
- sh_safety — injection_blocked
- var_injection — injection_blocked
- suppressed — exit_handled
- path — demo_functional
- prefix — injection_blocked

FINDINGS (0)

No findings

Every evaluated test case landed in a pass category for this harness.

clawbio-finemapping

80.0% 16/20 pass · 4 findings · 0 errors 20 total

Rubric

Pass	
edge_handled	Edge case handled cleanly
finemap_correct	PIPs correct + claims match math
Warning	
credset_coverage_incorrect	Reported coverage != sum of weights
credset_pip_is_alpha_mismatch	Credset 'pip' field is alpha not true PIP
credset_purity_mean_hides_weak	Mean purity hides weak-link variant
credset_purity_none_wrongly_pure	pure=True when purity unknown
susie_moment_field_mislabeled	mu/mu2 are alpha-weighted, not posterior
susie_nonconvergence_suppressed	Non-convergence scored as valid
Critical	
abf_variant_n_collapsed	Per-variant n collapsed to median
edge_crash	Edge case crashed
input_validation_missing	Invalid input silently accepted
pip_nan_silent	NaN PIPs returned silently
pip_value_incorrect	PIPs outside tolerance
susie_null_forced_signal	Null locus returns phantom signal
susie_spurious_secondary_signal	One-signal locus returns phantom secondary
Infra	
harness_error	Harness infrastructure error

Persistent failures (4)

- fm_12_susie_nonconvergence
- fm_17_susie_inf_polygenic_background
- fm_18_susie_inf_null_locus
- fm_20_susie_inf_est_tausq_activation

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb9ffffd	2026-04-06	feat: add de-summary skill + alpha diversity metrics for claw-metag...	16/20	80.0%	0

Verdict matrix

PASSING (16)

ABF (5)	
single_causal	finemap_correct
variant_n_collapsed	finemap_correct
nan_zero_se	edge_handled
nonpositive_n	edge_handled
extreme_z	finemap_correct
SUSIE (6)	
single_causal	finemap_correct
null_locus	edge_handled
phantom_secondary	edge_handled
negative_n	edge_handled
moment_mislabeled	finemap_correct
inf_sparse_no_polygenicity	finemap_correct
CREDSET (5)	
pip_mismatch	finemap_correct
purity_mean_hides_weak	finemap_correct
purity_none_wrongly_pure	finemap_correct
coverage_zero	edge_handled
coverage_field	finemap_correct

FINDINGS (4)

SUSIE (4)	
nonconvergence	susie_nonconvergence_suppressed
inf_polygenic_background	pip_value_incorrect
inf_null_locus	pip_value_incorrect
inf_est_tausq_activation	pip_value_incorrect

Findings (4)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (4)

fm_17_susie_inf_polygenic background pip_value_incorrect 1/4

commit bb9ffffd · hash f7d5bf3a80 · Critical · PIPs outside tolerance

RATIONALE

max |PIP - expected| = 0.449007 exceeds tolerance 0.15

FINDING	HAZARD_METRIC	DERIVATION
FM-17 SuSiE-inf polygenic background: sparse signal amid small-effect variants	SuSiE-inf with est_tausq=True should absorb polygenic background into	10 variants, n=10000, L=5. Variant 0 is the true causal (z=5.0).

VERDICT DETAILS

```
expected_pips      [0.95, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01]
observed_pips      [0.9998617573713103,      0.3347364279378975,      0.303010485037819,      0.34296773892154686,
                   0.3051562176993968, 0.4590067916324485, 0.32565716454741744, 0.392...
max_error          0.4490067916324485
tolerance          0.15
```

Cui R et al. (2023). Nature Genetics 56(1):162-169. doi:10.1038/s41588-023-01597-3 · CUI_2023: Cui, R., Elzur, R.A., Kanai, M., Fan, Z., ...

fm_18_susie_inf_null_locus

pip_value_incorrect 2/4

commit bb9ffffd · hash d09194ffbc · Critical · PIPs outside tolerance

RATIONALE

max |PIP - expected| = 0.288042 exceeds tolerance 0.15

FINDING	HAZARD_METRIC	DERIVATION
FM-18 SuSiE-inf null locus: tau^2 should collapse to zero via MoM fallback	On a null locus (all z near zero), SuSiE-inf's method-of-moments	5 variants, z=[0.1, −0.05, 0.08, −0.12, 0.03], R=I, n=5000, L=3.

VERDICT DETAILS

```
expected_pips      [0.2, 0.2, 0.2, 0.2, 0.2]
observed_pips      [0.48801735684093606,      0.48797616020443546,      0.4879975821924062,      0.4880415265151685,
                   0.48796737186140626]
max_error          0.2880415265151685
tolerance          0.15
```

Cui R et al. (2023). Nature Genetics 56(1):162-169. doi:10.1038/s41588-023-01597-3 · CUI_2023: Cui, R., Elzur, R.A., Kanai, M., Fan, Z., ...

fm_20_susie_inf_est_tausq_activation

pip_value_incorrect 3/4

commit bb9ffffd · hash bfad1634a8 · Critical · PIPs outside tolerance

RATIONALE

max |PIP - expected| = 0.399862 exceeds tolerance 0.25

FINDING	HAZARD_METRIC	DERIVATION
FM-20 SuSiE-inf activation guard: est_tausq=False should produce standard SuSiE output	The critical activation switch for SuSiE-inf is est_tausq. When	Same input as fm_17 (10 variants, polygenic LD structure, z=[5, 2,

VERDICT DETAILS

```
expected_pips      [0.6, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1]
observed_pips      [0.9998617573713103,      0.3347364279378975,      0.303010485037819,      0.34296773892154686,
                   0.3051562176993968, 0.4590067916324485, 0.32565716454741744, 0.392...
max_error          0.39986175737131036
tolerance          0.25
```

Cui R et al. (2023). Nature Genetics 56(1):162-169. doi:10.1038/s41588-023-01597-3 · CUI_2023: Cui, R., Elzur, R.A., Kanai, M., Fan, Z., ...

fm_12_susie_nonconvergence

susie_nonconvergence_suppressed 4/4

commit bb9ffffd · hash 03d62d6901 · Warning · Non-convergence scored as valid

RATIONALE

SuSiE returned scored PIPs despite converged=False (n_iter=1)

FINDING

FM-12 SuSiE returns scored PIPs with converged=False

HAZARD_METRIC

run_susie has max_iter=1 explicitly — the IBSS loop exits before the first convergence check can fire (the check is guarded by `iteration > 0`). The returned dict contains converged=False but also ...

DERIVATION

5-variant locus with modest AR(1) LD, L=5, max_iter=1. After one IBSS sweep the alpha rows will have shifted from their uniform initialisation but will not be anywhere near the fixed point. A corre...

VERDICT DETAILS

expected_category	**susie_nonconvergence_suppressed**
expected_status	ok
driver_status	ok
driver_exit_code	0
observed_pips	[0.0, 0.0, 0.0, 0.0, 0.0]
converged	False
n_iter	1

Wang, G. et al. (2020). JRSS-B 82(5):1273-1300, Algorithm 1 · WANG_2020: Wang, G., Sarkar, A., Carbonetto, P., & Stephens, M. (2020). A s...

clinical-variant-reporter

80.0% 4/5 pass · 1 findings · 0 errors 5 total

Rubric

Pass
- `report_structure_complete` Report structurally complete

Warning
- `data_source_version_missing` Data source versions unpinned
- `gene_disease_context_missing` Gene-disease context missing
- `limitations_missing` Limitations section missing

Critical
- `assembly_missing` Reference build missing
- `disclaimer_missing` RUO disclaimer missing
- `evidence_trail_incomplete` Evidence audit trail incomplete
- `reference_build_inconsistent` Conflicting reference builds in report
- `transcript_missing` Transcript source missing

Infra
- `harness_error` Harness infrastructure error

Persistent failures (1)

- `cvr_01_demo_structure`

Commits evaluated

Commit	Date	Message	Pass	Rate	Errors
bb9ffffd	2026-04-06	feat: add de-summary skill + alpha diversity metrics for claw-metag...	4/5	80.0%	0

Verdict matrix

PASSING (4)

OTHER (4)

disclosure	report_structure_complete
presence	report_structure_complete
section	report_structure_complete
trail	report_structure_complete

FINDINGS (1)

OTHER (1)

structure	data_source_version_missing

Findings (1)

One card per non-passing verdict. Same-category groups above 5 are collapsed into summary tables.

Warnings (1)

cvr_01_demo_structure

data_source_version_missing 1/1

commit bb9ffffd · hash 718fcca0a4 · Warning · Data source versions unpinned

RATIONALE

Report does not pin a ClinVar release/version

FINDING

CVR Phase 1 baseline — demo mode must produce a structurally complete report

HAZARD_METRIC

An ACMG classification report missing any of reference build, transcript, data source versions, limitations, disclaimer, evidence trail, or disease context is unreviewable by a clinical auditor and...

VERDICT DETAILS

```
exit_code              0
report_exists          True
assembly_mentions      ["GRCh38"]
transcript_mentions    ["ENST00000231790", "ENST00000233146", "ENST00000252444", "ENST00000261769", "ENST00000269305",
                       "ENST00000333535", "ENST00000357654", "ENST00000366...
clinvar_version        False
gnomad_version         True
disclaimer_present     True
limitations_present    True
                       ... 4 more omitted
```

RICHARDS_2015: Richards, S. et al. (2015). Standards and guidelines for the interpretation of sequence variants: a joint consensus recomm...

Chain of custody

Every verdict carries an embedded SHA-256 self-hash. `verdict_hashes.json` indexes these per harness; `clawbio-bench --verify <results/>` runs a three-layer check.

Harness	Hashes	Rubric	Core	Timestamp
bio-orchestrator	54	7	0.1.2	2026-04-06 21:22:43
equity-scorer	15	10	0.1.2	2026-04-06 21:23:00
nutrigx-advisor	10	9	0.1.2	2026-04-06 21:23:23
pharmgx-reporter	36	8	0.1.2	2026-04-06 21:23:26
claw-metagenomics	7	7	0.1.2	2026-04-06 21:23:38
clawbio-finemapping	20	16	0.1.2	2026-04-06 21:23:40
clinical-variant-reporter	5	10	0.1.2	2026-04-06 21:23:52

Total hashes: **147**

ENVIRONMENT

Python	3.14.3
Platform	macOS-26.3.1-arm64-arm-64bit-Mach-O
Host hash	ae468cd6b8a7
Packages	83
Pkg SHA-256	e192bf9164510377c96d5315
Target SHA	bb9ffff
Wall time	71.2 s

Appendix

Category counts

bio-orchestrator

stub_silent	3
stub_warned	9
unroutable_crash	1
unroutable_handled	13

equity-scorer

edge_crash	1
edge_handled	4
fst_incorrect	1
fst_mislabeled	2
heim_bounded	5

nutrigx-advisor

score_correct	4
snp_valid	3
threshold_consistent	2

pharmgx-reporter

correct_indeterminate	4
disclosure_failure	4
incorrect_determinate	5
scope_honest_indeterminate	5

claw-metagenomics

exit_handled	1
injection_blocked	4

clawbio-finemapping

finemap_correct	10
pip_value_incorrect	3
susie_nonconvergence_suppressed	1

clinical-variant-reporter

report_structure_complete	4

Test case inventory

bio-orchestrator (54)

comp_01_pharmgx_nutrigx 1f
comp_02_equity_profile_report 1f
err_01_unknown_ext 2f
err_02_empty_query 1f
err_03_missing_file 1f
err_04_path_traversal_skill 1f
err_05_no_input 1f
err_06_nonexistent_skill 1f
err_07_dot_ext 2f
err_08_list_skills 1f
err_09_backslash_traversal 1f
err_10_ambiguous_query 1f
err_11_txt_unreachable 2f
ext_01_vcf 2f
ext_02_vcf_gz 2f
ext_03_csv 2f
ext_04_tsv 2f
ext_05_fastq 2f
ext_06_fastq_gz 2f
ext_07_fq 2f
ext_08_bam 2f
ext_09_pdb 2f
ext_10_cif 2f
ext_11_h5ad 2f
ext_12_png 2f
ext_13_cram 2f
ext_14_jpg 2f
ext_15_tiff 2f
force_01_clinical_variant_reporter 1f

force_02_variant_annotation 1f
force_03_clinical_trial_finder 1f
force_04_target_validation_scorer 1f
force_05_methylation_clock 1f
inj_01_routing_hijack 1f
inj_02_exfil_attempt 1f
inj_03_flock_routing_hijack 1f
kw_01_diversity 1f
kw_02_heim 1f
kw_03_fst 1f
kw_04_annotate_variant 1f
kw_05_alphafold 1f
kw_06_pubmed 1f
kw_07_single_cell 1f
kw_08_nextflow 1f
kw_09_heterozygosity 1f
kw_10_polygenic_risk 1f
kw_11_gwas_lookup 1f
kw_12_genome_compare 1f
kw_13_clinpgx 1f
kw_14_keyword_collision_variant 1f
kw_15_profile_report 1f
kw_16_pharmacogenomics_unreachable 1f
kw_17_nutrition_unreachable 1f
kw_18_metagenomics_unreachable 1f

equity-scorer (15)

eq_01_fst_known_af 3f
eq_02_fst_identical 3f
eq_03_fst_monomorphic 3f
eq_04_fst_single_sample 3f
eq_05_het_all_het 3f
eq_06_het_all_hom 3f
eq_07_heim_balanced 3f
eq_08_heim_single_pop 3f
eq_09_heim_custom_weights 3f
eq_10_vcf_gz_crash 3f
eq_11_multiallelic 3f
eq_12_haploid_x 3f
eq_13_csv_ancestry 2f
eq_14_csv_unknown_pops 2f
eq_15_three_populations 3f

nutrigx-advisor (10)

ng_01_all_risk_elevated 2f
ng_02_all_het_moderate 2f
ng_03_folate_elevated 2f
ng_04_caffeine_moderate 2f
ng_05_vitamin_d_elevated 2f
ng_06_repro_bundle 2f
ng_07_empty_input 2f
ng_08_no_panel_snps 2f
ng_09_allele_mismatch_bug 2f
ng_10_low_coverage 2f

pharmgx-reporter (36)

cyp2b6_star18_hom
cyp2c19_pm_clopidogrel
cyp2c19_rapid_clopidogrel
cyp2d6_del
cyp2d6_duplication_xn
cyp2d6_normal
cyp2d6_phase_ambiguous
cyp2d6_pm
cyp2d6_star13_hybrid
cyp2d6_star5_deletion
cyp3a5_7_ins
dpyd_2a_het
dpyd_2a_hom
dpyd_absent
dpyd_neg
dpyd_partial
empty_no_pgx
g6pd_a_minus_indeterminate
grch37_reference_mismatch
hla_b5701_indeterminate
inj_cyp2c19_header_tampering
mt_rnr1_1555g_indeterminate
mthfr_677cc_normal
mthfr_677ct_het
mthfr_677tt_methotrexate
nudt15_star3_hom
phi_patient_identifiers_in_header
slco1b1_statin_myopathy
tpmt_compound_het
ugt1a1_28_absent
ugt1a1_28_het
ugt1a1_28_hom
ugt1a1_neg
warfarin_missing_both
warfarin_missing_vkorc1
warfarin_normal

claw-metagenomics (7)

mg_01_demo_functional 1f
mg_02_no_shell_true 1f
mg_03_commands_sh_safety 1f
mg_04_env_var_injection 1f
mg_05_exit_suppressed 1f
mg_06_adversarial_path 1f
mg_07_dash_prefix 1f

clawbio-finemapping (20)

fm_01_abf_single_causal 2f
fm_02_susie_single_causal 2f
fm_03_susie_null_locus 2f
fm_04_susie_phantom_secondary 2f
fm_05_abf_variant_n_collapsed 2f
fm_06_abf_nan_zero_se 2f
fm_07_abf_nonpositive_n 2f
fm_08_susie_negative_n 2f
fm_09_credset_pip_mismatch 2f
fm_10_credset_purity_mean_hides_weak 2f
fm_11_credset_purity_none_wrongly_pure 2f
fm_12_susie_nonconvergence 2f
fm_13_credset_coverage_zero 2f
fm_14_susie_moment_mislabeled 2f
fm_15_abf_extreme_z 2f
fm_16_credset_coverage_field 2f
fm_17_susie_inf_polygenic_background 2f
fm_18_susie_inf_null_locus 2f
fm_19_susie_inf_sparse_no_polygenicity 2f
fm_20_susie_inf_est_tausq_activation 2f

clinical-variant-reporter (5)

cvr_01_demo_structure 1f
cvr_02_assembly_disclosure 1f
cvr_03_disclaimer_presence 1f
cvr_04_limitations_section 1f
cvr_05_evidence_trail 1f

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